

Implications of Using the Herfindahl-Hirschmann Index (HHI) For Purposes Other Than Concentration

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The Herfindahl-Hirschman Index (HHI) is one of the more commonly used measures in the Management and Economics literatures. While its original use was measuring market concentration, it has been extended well beyond that. Already in the first six months of 2025 forty-one papers using HHI have been published in the top five strategy journals, the majority of which use HHI for purposes other than concentration. The measure has three properties that potentially affect empirical inference: 1) it is “non-injective”—any given HHI represents an infinite set of distributions, 2) it is a compound measure--intentionally combining two dimensions (count and dispersion) which are substitutes for capturing concentration, but may not be substitutes for other constructs, and 3) it is asymptotic. We assess whether these properties jointly affect inferences in strategy research. To do so, we replicate a prior study which employs HHI to test the impact of geographic diversification on firm value. We find that results with HHI are not robust across samples and specifications. We therefore recommend researchers abandon the use of HHI for applications other than concentration, unless they can demonstrate theoretically that count and dispersion are substitutes in their application.

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I. Introduction

One of the more commonly used measures in the strategy and economics literatures is the Herfindahl-Hirschman Index (HHI). Its origins date to 1945, when Hirschman introduced it in the book, “National Power and the Structure of Foreign Trade” to measure market concentration. A very similar measure was reinvented by Herfindahl (1950) for the same purpose, and thus the measure has the compound name.

HHI entered the strategy literature when Song (1982) employed it to characterize the level of firms’ diversification. Since that time, both the frequency and breadth of its use in the top five strategy journals have grown steadily, as shown in Figure 1. Already in the first six months of 2025, 41 such articles used HHI. While the dominant use continues to be concentration (both market concentration and ownership concentration), that still comprises less than 50% of all uses. The next most prevalent use is diversification, followed by technological breadth and team composition.

[Insert Figure 1 Here]

The measure has three properties that potentially affect empirical inference: 1) it is “non-injective” 2) it is a compound measure and 3) it is asymptotic. We discuss each of these properties in turn.

Non-injective. An infinite set of distributions share the same HHI. Most notably, for any value of HHI, there is both a uniform distribution and distributions which are highly skewed. This inability to map a given HHI to a particular distribution makes causal inference impossible.

Compounded. HHI deliberately combines “unequal distribution” and “fewness” because Hirschmann (1964) viewed the two dimensions as substitutes in capturing market power. Indeed,

recent theoretical work indicates that at least under Cournot competition, the two are substitutes (Spiegel 2020).

It's not clear these two dimensions are substitutes in other applications of HHI, however. In fact, the compound nature of HHI may inhibit insight. Take for example, the use of HHI to measure corporate diversification. The compound nature makes it difficult to interpret what an increase in diversification actually means. More problematic is the possibility that the two dimensions of HHI may have different effects on firm value.

Asymptotic. The final concern with HHI is that it is asymptotic. Because it is bounded by 1, a firm diversifying at a constant rate will appear to be decreasing its level of diversification over time. This is a concern in empirics examining whether firms are becoming more or less diversified.

Our goal is to determine whether these properties of HHI jointly affect empirical inference. To do so, we replicate a prior study, Goetz, Laeven, Levine (2013) (GLL) which utilizes the HHI, plus three other measures, to examine the impact of diversification on firm value. While diversification is not the only non-concentration use of HHI within strategy, it is the second most prevalent use. Moreover, because technical diversity and team diversity are similar to corporate diversification, it is likely that if results are sensitive to use of HHI for diversification purposes, it is likely they are sensitive to its use in these other applications as well.

We begin by replicating GLL's result that diversification decreases firm value. We then use the authors' raw data, which they generously shared, to construct three additional measures of diversification: count, entropy, and dispersion. We construct all seven measures (four from GLL and three new ones) for the firms in our setting, then present pairwise comparisons of the measures to illustrate the extent to which they differ. We then employ the empirical model in GLL to test the relationship between each measure and firm value across a number of specifications and sample

constructions. While our primary interest is in HHI, the broader set of measures may allow us to identify suitable alternatives.

We find first, that the coefficients HHI change sign across specifications. Thus, use of HHI does indeed affect inferences. Prior results with HHI concluding that diversification either increases or decreases performance are unlikely to be robust to specifications or sample restrictions.

We find second, that coefficients on count are consistently positive in explaining firm value, while coefficients on dispersion are consistently negative. This is true in univariate fixed effects regressions, in regressions which combine count with other measures, and in instrumental variable regressions, where we instrument for count using state banking deregulations. Thus decomposing HHI into its two constituents leads to reliable results (as well as greater insight).

Taken together, the results indicate that the idiosyncratic properties of HHI that make it suitable for studies of market concentration, make it unsuitable for at least one important application—diversification. Thus, we recommend abandoning HHI for that application, and instead replacing it with its two constituents: count and dispersion. For other applications, we recommend that researchers provide a theoretical rationale for why count and dispersion would be substitutes before using HHI in those applications.

The paper proceeds as follows. First, we discuss diversification. Next, we discuss the HHI measure, followed by a discussion of alternative diversification measures. Fourth and fifth, respectively, we present our empirical approach and results. Finally, we close with a discussion of implications.

II. What is Meant by Diversification

Diversification has long occupied a central place in strategy research. Penrose (1959) pioneered the idea that when faced with declining opportunities in their current market, firms could grow by

diversifying into new markets that exploited slack resources. Chandler (1962) extended her thinking to define the organizational structure best suited to implement this resource-based growth. In particular, he advocated a multi-divisional form, with centralized control over strategic decisions, but decentralized control of operational decisions. Rumelt (1974) empirically tested the Penrose/Chandler view by first characterizing firms according to their organizational structure: single business, dominant-vertical, dominant-constrained, dominant-linked, dominant-unrelated, related-constrained, related-linked and unrelated. He next assessed the performance of firms with each structure, then compared across structures. He found that firms with related diversification structures had the highest performance, followed by those with a single business. Firms with unrelated diversification structures had the worst performance. These findings reinforced the Penrose/Chandler view that growth through diversification was valuable, but only when the slack resources were deployed in related settings.

While Rumelt utilized discrete measures of diversification (the eight structures), their construction required detailed analysis of firms' operations. Since then, strategy scholars have sought measures that were continuous, such that they could capture the degree of diversification. Montgomery (1982) for example, used a slight variant of HHI to capture the categorical measures in Rumelt 1974. For all but two of the Rumelt categories, she found the correlations with the means of the continuous measure to be highly significant.

While the Penrosian motive for diversification is the most prevalent in strategy, it is worth noting that diversification can also add value by risk-pooling, so as to minimize susceptibility to adverse changes in any given market. The risk pooling motive for diversification across markets mimics the motivation for diversification in investment portfolios. Markowitz (1952) argued that investors could achieve a given return at lower risk than that for any given stock, by choosing a portfolio of

stocks for which the weighted sum of pairwise covariances in returns was minimized. Thus, for any level of tolerable risk, an investor should be able to construct a portfolio whose returns are higher than those of a single stock. Conversely, for any given level of returns, there should be a portfolio whose risk is lower than a single stock offering those returns.

Note that the two motives for diversification, resource exploitation and risk pooling appear to offer conflicting prescriptions. The resource exploitation motive implies greater performance when units are positively related; the risk-pooling motive implies greater performance when units are negatively related. This is not strictly true, of course. A firm's business units can be related along one dimension, while negatively correlated along another. The global cement company, Cemex, for example, is geographically diversified to minimize vulnerability to political and currency risk. However, operationally, all units produce the same commodity, which allows them to share centralized services, as well as technological and other forms of expertise.

Because these motives for diversification are so different, it's unlikely a single measure is appropriate for all forms of diversification. Indeed Gort (1962) advised that the "*choice of measure depends upon the problem one chooses to examine. To predict responses of output to fluctuations in the firm's primary industry, then you want a measure of how concentrated operations are in that industry. Conversely, if you want to predict the likelihood of entering a more attractive industry, then a count of industries may be sufficient*" (Gort, 1962:10).

Accordingly, Gort tested three measures. The first measure was based on the share of payroll in the firm's primary industry; the second was based on the number of industries in which the firm produces; the third was a composite of the first two. This composite measure was the forerunner to HHI for measuring diversification. Gort used all three measures to characterize the level of diversification in a longitudinal panel of 111 large firms and a cross-section of 721 firms. The

correlation between the share measure and the composite measure was 0.94, though Gort preferred the composite measure, because it was more stable across industry definition, e.g., 2-digit versus 4-digit SIC.

III. The Herfindahl-Hirschman Index (HHI)

What has come to be known as the Herfindahl-Hirschman Index (HHI) was developed by Herfindahl in his dissertation, “Concentration in the Steel Industry (1950).¹ The goal of the dissertation was to determine if the steel industry had become more or less monopolistic over time. To assess that, Herfindahl required a measure of monopoly power.

There were two approaches to measuring monopoly power at the time: measures of de facto power (such as the Lerner’s (1934) index of price-cost margins), and concentration measures such as Crowder’s (1937) percent of output controlled by the largest 4 firms in an industry (now referred to as the C-4 concentration ratio). While the Lerner measure “*is a neat measure of the degree of monopoly power*” (Herfindahl, 1950:22), reliable marginal cost data is rarely available. Thus, Herfindahl sought a measure constructed from readily available data that improved upon Crowder.

His two concerns with the Crowder measure were 1) it ignores the number of firms in the industry, and 2) it doesn’t convey the extent of domination by the leading firms, which Herfindahl equated with inequality. Herfindahl felt the number of firms had an obvious bearing on the ability to collude, and while Crowder took the number of firms into account in his study, he did so separately from concentration. Herfindahl wanted a single measure of market power that captured both the number of firms and the distribution of output across them.

¹ The measure became the Herfindahl-Hirschman index when it was discovered that Hirschman (1945) had earlier developed an almost identical measure for the same purpose. Hirschman (1964) describes the paternity of the index.

His measure, C , which here forward we refer to as HHI, is computed as the sum of the squares of firms' market shares:

$$C = \sum_{i=1}^m \left(\frac{x_i}{\sum_{i=1}^m x_i} \right)^2 = \sum_{i=1}^m (s_i)^2 \quad (1)$$

where m is the number of firms, x_i is the output of the i^{th} firm, and s_i is its share of industry output.

One concern with HHI is that for any given value, there is an infinite variety of distributions that it represents. Herfindahl recognized this, and in fact constructed a table to show hypothetical distributions all sharing the same HHI. For example, an industry of two firms, each with 50% shares, has the same HHI (0.5) as an industry of five firms, with shares of 65%, 27%, 6.5%, 1.0% and 0.5%. (The corresponding C-4 measures would be 100% and 99.5%).

Relatedly, for any given HHI, there exists a uniform distribution with that HHI (though its count is not always an integer). In particular, the number of equal-sized firms that generate a given HHI is $1/\text{HHI}$ (Adelman 1969). Thus, the same HHI can capture a uniform distribution as well as a highly skewed distribution, as in Herfindahl's example.

A second, perhaps more important concern, is the compound nature of HHI. While count and dispersion appear to be substitutes in gauging market power, it's unclear they are substitutes in other applications.

IV. Use of HHI to Measure Diversification

The first use of HHI to measure diversification appears to be Berry (1971). Like Gort, he recognized that diversification should be gauged not just by the number of industries, but by the distribution of activity across industries, and cites Gort for that observation. Berry employed "a more complete index of diversification", HHI, without discussing why HHI is more complete than Gort's compound measure. He then compared changes in the level of firms' diversification over

time. He found that when using the count measure there had been a 48% increase in firms' diversification, but with HHI, there had only been a 2.8% increase.

This is an artifact of the third concern with HHI. It is asymptotic. Because it is bounded by 0 and 1, a firm diversifying at a constant rate will appear to decrease its level of diversification over time. For example, a firm which starts with three equal-sized units and grows that number by 20% each year, will have a diversification rate in year 15 that is less than half what it was in year 1 (as shown in Figure 2). This is a potential concern in empirics examining whether firms are becoming more or less diversified over time.

[Insert Figure 2 Here]

Another artifact of the asymptote problem occurred when Berry switched from a 4-digit to a 2-digit definition of industry. There he found the rate of diversification doubled when using HHI, but decreased by 60% when using count (relative to the 4-digit rates).

V. Alternative Measures of Diversification

While HHI is the most common measure of diversification, there are others. Perhaps the most straightforward and long-lived measure of diversification is a simple count of a firm's business units (BU). These BUs can be defined by industry, as in Gort (1962) and Rumelt (1974), or by geographic market, as in Jacquemin and Berry (1979). While count captures the fewness aspect of diversification, it does not capture the dispersion across BUs.

This concern led to a commonly employed alternative to HHI, called the entropy measure (Jacquemin and Berry, 1979). Entropy is constructed in similar fashion to HHI—with a goal of expressing both fewness and distribution of assets across units in a single continuous index. Not surprisingly, the entropy measure is highly correlated with HHI. Entropy is constructed as:

$$E = \sum_{i=1}^n s_i \ln \left(\frac{1}{s_i} \right) \quad (2)$$

where s_i denotes the i th subunit's share. Here a subunit's share is weighted by the logarithm of its inverse. Accordingly, while the value of HHI is principally influenced by large subunits, the opposite is true for entropy, it is principally influenced by small units. This difference notwithstanding, entropy is vulnerable to the two principal concerns we highlight for HHI—it is a compound measure, and it is not unique. There is an infinite variety of distributions for any given value of entropy.

One additional measure of a distribution that seems relevant to diversification, though to our knowledge has not been used for that purpose: is the standard deviation, σ . What distinguishes σ from HHI and entropy is that it captures dispersion, while ignoring count. Firms with larger values of σ exhibit greater differences between units. To minimize the impact of scale, we normalize σ by the mean of the distribution, $\bar{x}$, to form the coefficient of variation, CV , given in Equation 3:

$$CV = \frac{1}{\bar{x}} \sqrt{\frac{\sum_{i=1}^n (x_i - \bar{x})^2}{(n-1)}} \quad (3)$$

where n is the number of units, x_i is the firm's assets in unit i , and $\bar{x}$ is the firm's mean assets across all its units. Note this computation is only feasible for firms with at least two subunits (else the denominator is zero). Of course below this level, firms aren't diversified.

To provide better intuition for the measures, we do two things. First, we provide a series of pairwise comparisons of the measures across all firms in our sample in Figure 2. These comparisons indicate that there is very little similarity between measures. Thus choice of measures is likely to affect inferences.

[Insert Figure 3 Here]

In addition to the pairwise comparisons for the entire sample, we present the evolution of each measure for the firm in our sample with the greatest number of units (Figure 4b). For reference, Figure 4a shows the introduction and evolution in scale for each subsidiary within the firm. We don't display count (n) because each line in Figure 3a represents a unit, so the set of lines capture count. What you can see by comparing Figures 3a and 3b is that while entropy seems to track the growing diversity, HHI suggests there is little difference between the diversification in 1994, when there were five units and 2007, when there were 54 units. This stems from the asymptote property of HHI, that was depicted in Figure 3.

[Insert Figure 4 Here]

Looking next at CV, we see it is highly sensitive when there are relatively few units (1991-1998), but thereafter indicates the firm is maintaining fairly constant breadth in unit size. The CV is less than 1, indicating the standard deviation of unit size is below the mean.

V. Empirical Approach

We now examine whether use of HHI affects empirical inferences in strategy research.

A. Precedent paper

To assess whether the differences affect empirical inferences, we begin by replicating a prior paper, Goetz, Laeven, and Levine (2013) (GLL). GLL employs quarterly data on U.S. bank holding companies (BHC), to test the impact of geographic diversification on firm value using the following model:

$$q_{ist} = \beta D_{ist} + X'_{ist} \rho + \delta_i + \delta_t + \varepsilon_{ist} \quad (4)$$

where q_{ist} (measured as *Tobin's q*), denotes firm value of BHC i , headquartered in state s , during quarter t . D_{ist} denotes the level of geographic diversification, captured using alternative measures, X'_{ist} is a matrix of conditioning information, and δ_i and δ_t are BHC and quarter fixed effects, respectively.

Because GLL have concerns similar to ours, they employ four alternative measures of diversification: (1) a *diversification dummy* equal to 1 if the BHC has subsidiaries (banks) in more than one state, and zero otherwise; (2) the *fraction of assets held out-of-state*; (3) the *HHI* of the BHC's assets across states, inverted to $1 - \text{HHI}$, so that higher values correspond to higher levels of diversification; and (4) *Mean distance*, measured as the natural logarithm of the average distance between a BHC's headquarters and its subsidiaries. While we discuss the GLL results in greater detail in our replication, their main finding is that across all four measures, geographic diversification reduces BHC value. The authors interpret this as evidence that diversification intensifies agency problems within BHCs.

B. Addressing firm choice to diversify

Equation 4 is a firm fixed effects specification. Thus, in principle it ties changes in firm value to changes in levels of diversification. Having said that, there may be firm characteristics that jointly drive firm value and diversification, and thus firm value increases (or decreases) along with diversification, but not because of it. We next address that concern.

Probably the most common approach to addressing selection effects is to match “treatment firms” (in our case diversified firms) to “control firms” (non-diversified firms) that are similar other than for the treatment, e.g., Villalonga (2004). This approach requires a discrete treatment, such as whether a firm is diversified. Because we are assessing continuous measures,

this approach is not available to us.

A second approach to dealing with selection effects is to employ a difference-in-differences (DiD) model, which estimates the treatment effect of a discrete event by comparing outcomes post-event to the firm's prior outcomes (pre-trend) as well to outcomes of untreated firms. GLL use this approach to deal with selection effects. In particular, they examine firm value ten quarters prior to and ten quarters following the quarter in which a firm first establishes an out of state subsidiary. They find there is no pre-trend in *Tobin's Q* prior to diversification, but that *Tobin's Q* decreases in each quarter thereafter. This is fairly compelling evidence that diversification is causing the decrease in firm value. As with matching models however, the DiD approach is infeasible for us, because we are assessing continuous measures of diversification, so don't have discrete events..

Accordingly, we adopt GLL's other strategy for dealing with selection effects, which is to use an instrumental variable (IV) which should have no influence on firm value other than through its effect on diversification. The IV is time since the BHC's headquarters' state enacted laws allowing out of state BHCs to hold in-state banks. Note that while interstate acquisition was enabled nationally in 1994 by the Riegle-Neal Interstate Banking and Branching Act, prior to that time, bank acquisition was governed by the Bank Holding Company Act of 1956. The BHC Act prohibited BHC acquisition of out-of-state banks *unless specifically authorized* by laws in the target bank's state. While no states had such laws in 1956, nearly all states adopted interstate banking laws between the early 1980s and the passage of Riegle-Neal. Timing of laws in the BHC state may appear to be an odd instrument, given that the laws pertain to the target state. However, in practice the state laws authorizing acquisitions in the target state, required reciprocal laws in the acquiring state. Accordingly, laws allowing out-of-state acquisitions in the

BHC's state are equivalent to ones allowing BHC's to acquire banks in the target state.

We follow GLL in utilizing state deregulation as our IV. In the first stage, we model the *count* of states in which a BHC has subsidiaries as a function of time since the BHC headquarters' state deregulated banking. In the second stage, we model the impact of predicted *count* and other measures of diversification on firm value. While econometrically, it would be feasible to use any of the diversification measures in the first stage, to solve the selection problem, we want a measure which reflects the choice firms are making in response to deregulation. Of our diversification measures, the most obvious choice was *count* of states. This most closely reflects a decision to further diversify. It seems less likely firms are attempting to optimize *mean distance*, *HHI*, *entropy*, or *dispersion*.

C. Data

The GLL dataset, which we employ, is drawn from two sources. The first source is the Consolidated Financial Statements for Bank Holding Companies (FR Y-9C report), collected by the Federal Reserve from domestic BHCs. These reports provide quarterly data obtained from balance sheets, income statements, and detailed supporting schedules. The second source is the Reports of Condition and Income (Call Reports) available through the Federal Financial Institutions Examination Council (FFIEC), also collected on a quarterly basis. While the FR Y-9C report is at the BHC level, Call Reports are at the bank level (BHC subsidiary). GLL were generous in sharing the data used for their tests, as well as the code for matching the FR Y-9C and Call Report data.

Following GLL, we merged the banking data with information on stock prices from the Center for Research in Security Prices (CRSP) and calculated *Tobin's q* as the sum of market

capitalization, total liabilities, and perpetual preferred stock divided by the value of total assets.

While we began with the GLL dataset, we needed to add subsidiary data to form the new diversification measures. These data included detailed information on 125,216 subsidiary-quarter observations for 993 unique BHCs, covering the same period as the GLL. With this new file we generated the three new diversification measures and replicated the existing four measures in GLL, then aggregated the data to the BHC-quarter level. Following GLL, we omitted observations with outlier values for *Tobin's q* (the bottom and top 1% of the distribution). Once aggregated, the “recreated dataset” comprised 31,830 BHC-quarter for 950 unique BHCs (versus 992 in the GLL dataset) from the third quarter of 1986 to the fourth quarter of 2007. Summary statistics for the data are provided in Table 1.

[Insert Table 1 Here]

VI. Results

A. Replication of GLL

Results for tests of equation 4 for the original GLL measures along with the new measures are reported in Table 2. Models 1-4 comprise the replication of GLL. Results for the replication match those in GLL, indicating we have faithfully reproduced their data and empirical estimation.² As in GLL, we find a negative relationship between geographic diversification and firm value for each of their measures.

The remaining models in Table 2 introduce the alternative diversification measures: count, entropy and dispersion (CV). The coefficient estimates indicate that firm value increases with the

² Note we cluster standard errors at the BHC, while GLL cluster at the state-quarter. As a consequence, while are coefficient estimates are identical, their coefficients are reported as significant, while ours are not.

count of states in which the BHC is active, as well as with entropy. However, it decreases with dispersion. Thus, use of HHI affects empirical inferences in at least one form of strategy research—diversification. Use of HHI leads to a conclusion that diversification decreases firm value, while use of entropy leads to a conclusion that diversification increases firm value.

[Insert Table 2 Here]

To further explore the sensitivity of results to choice of diversification measure, we restrict the set of observations to those BHC-quarters in which the BHC is operating in at least two states (the number required to form *CV*). Only 21% of observations meet this criterion. This balanced panel allows us to see if changes in coefficient estimates across measures are due to sample restrictions necessitated by some measures, versus due to the measures themselves. Results with this panel are reported in Table 3. The coefficients on all the measures other than dispersion, are now positive. This likely reflects the fact that at this level of diversification, each of the measures is largely registering increases in extent of diversification (number of states), rather than dissimilarity across states. The coefficient for dispersion continues to be negative, and maintains its magnitude, but becomes insignificant because of the smaller sample size. Most importantly, HHI is now positive and significant. Thus, use of HHI is sensitive to sample construction. In this case the sensitivity appears to be to exclusion of observations where $m=1$.

[Insert Table 3 Here]

Because count and dispersion appear to have competing effects, we conduct an additional test, where we replicate Table 2, but add *count of states* to each model. This allows us to see the relative contributions of the two components of diversification that Hirschmann intentionally combined: unequal distribution and fewness. These results are reported in Table 4. In all models, the

coefficient for *count of states* is positive and significant, while the coefficients for all other diversification measures are negative. Since HHI and entropy are compound measures, a loose interpretation of their coefficients when controlling for count, is that they reflect the residual effects of dispersion

[Insert Table 4 Here]

The results across Tables 2-4 indicate that the relationship between diversification and firm value is actually twofold: *Tobin's q* is consistently and significantly increasing in the *count of states* in which a BHC is active but decreasing in all measures of dissimilarity. This strategically important nuance is obscured when using compound measures, such as HHI and entropy. More importantly, the results suggest that HHI is not useful as a measure of diversification—count and dispersion aren't substitutes for capturing diversification as they are for capturing concentration.

B. Controlling for Selection Effects

The results in Tables 2-4 are derived from firm fixed effects specifications. Thus, in principle they relate changes in firm value to changes in levels of diversification. However, there may be firm characteristics that jointly drive firm value and diversification. We address this concern using an instrumental variable (IV) approach. In the first stage (Table 5b), we model the *count of states* in which a BHC has subsidiaries as a function of the time since the headquarters' state deregulated banking and other controls. In the second stage (Table 5a), we model the impact of predicted *count* and other measures of diversification on firm value.

Looking first at stage 1, we see that deregulation is significant in predicting entry into other states. While firm performance (operating income and growth in operating income) is also important, these variables are less significant than deregulation. Looking next at stage 2, we see

that results match those in Table 4, and in fact appear stronger after controlling for the choice to diversify. Firm value is increasing in count but decreasing in all other measures of diversification when controlling for count.

[Insert Table 5 Here]

Thus, the main results as well as those from efforts to address selection effects, indicate that results with HHI are not robust. More importantly, the results indicate that the compound nature of HHI is inappropriate for measuring diversification, because the two components, count and dispersion are not substitutes for diversification as they are for market concentration. While increasing *count* increases firm value, that value decreases in the level of dissimilarity, at least in this setting and for this form of diversification.

VII. Discussion

The Herfindahl-Hirschman Index (HHI) is one of the more commonly used measures in the strategy and economics literatures. While its original use was measuring market concentration, it has been extended beyond that. The measure has three properties that potentially affect empirical inference: 1) it is “non-injective”—any given HHI represents an infinite set of distributions, 2) it is a compound measure--intentionally combining two dimensions (count and dispersion), and 3) it is asymptotic. Our goal was to assess whether these properties jointly affect inferences in applications other than concentration. To do so, we replicated an existing test using HHI to examine the impact of geographic diversification on firm value. We then expanded the replication to include additional diversification measures, a number of sample restrictions, as well as a decomposition of the compound measures into its two components: count and dispersion.

This set of tests produced three notable insights. First, we found that results with HHI changed

sign when going from the full sample to a balanced panel that drops observations with only one unit. Second, we found that decomposing HHI into its count and dispersion components led to more reliable results. In particular, count was positive in all specifications, while dispersion was negative in all specifications, though neither was reliably significant. These results were robust to efforts to control for selection effects through use of an instrumental variable. Thus, HHI appears unsuitable for measuring diversification.

There are caveats to our results and conclusions, of course. First, we look only at one application of HHI in strategy research: diversification. However given our findings of when and why they were problematic for diversification, we can infer when they will be problematic for other applications.

Looking first at market concentration. Theoretical work indicates that at least under Cournot competition, fewness and skewness are substitutes (Spiegel 2020). Therefore, combining them in a single measure is valid. Indeed, this was the logic of both Hirschman and Herfindahl when creating the HHI measure.

Looking next at ownership concentration, typically the issue is governance incentives—the greater the ownership share, the higher the incentives. This suggests that the concentrated owners share a common goal, just as concentrated firms share a common goal, which might make HHI suitable here as it is for market concentration, but this is an empirical issue.

Finally looking at the other two prevalent uses of HHI: technological breadth and team diversity, these appear similar to diversification, in that count and dispersion likely have differing effects. In the case of technology, more patents would seem to be better, but the patents are probably most valuable if they share a common core such that they can exploit shared firm resources. As for team diversity, one can imagine there might be an optimal team size, just as there may be an

optimal level of dissimilarity. But again, these are empirical questions, and ones that cannot be addressed using HHI.

In short, our result that HHI is not robust across specifications and sample restrictions suggests researchers abandon the use of HHI for applications other than concentration, unless they can demonstrate theoretically that count and dispersion are substitutes in their application.

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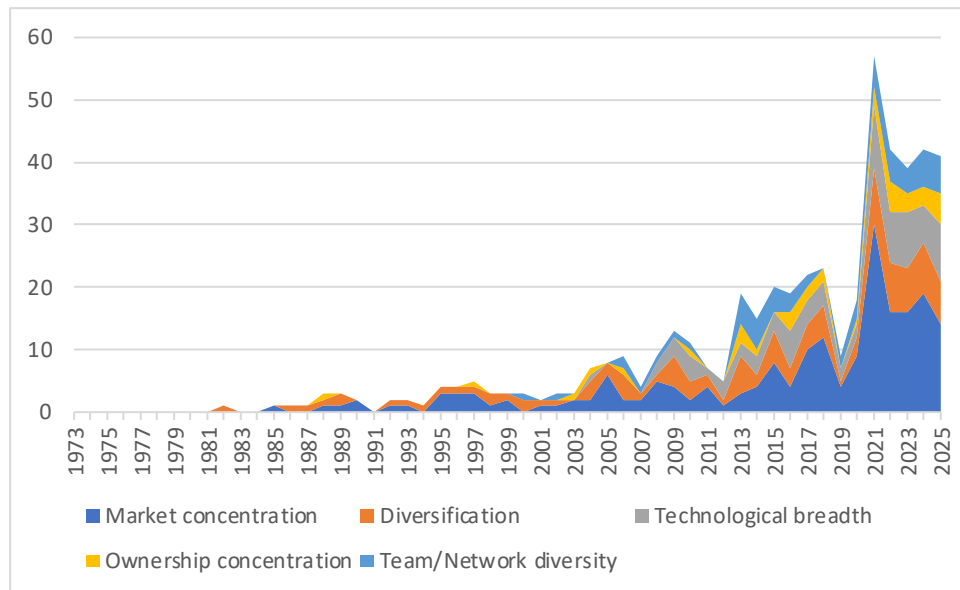


FIGURE 1. HERFINDAHL MAJOR USES ACROSS TIME, 1980-2025

(IN MAIN STRATEGY OUTLETS: SMJ, MS, OS, SS AND AMJ)

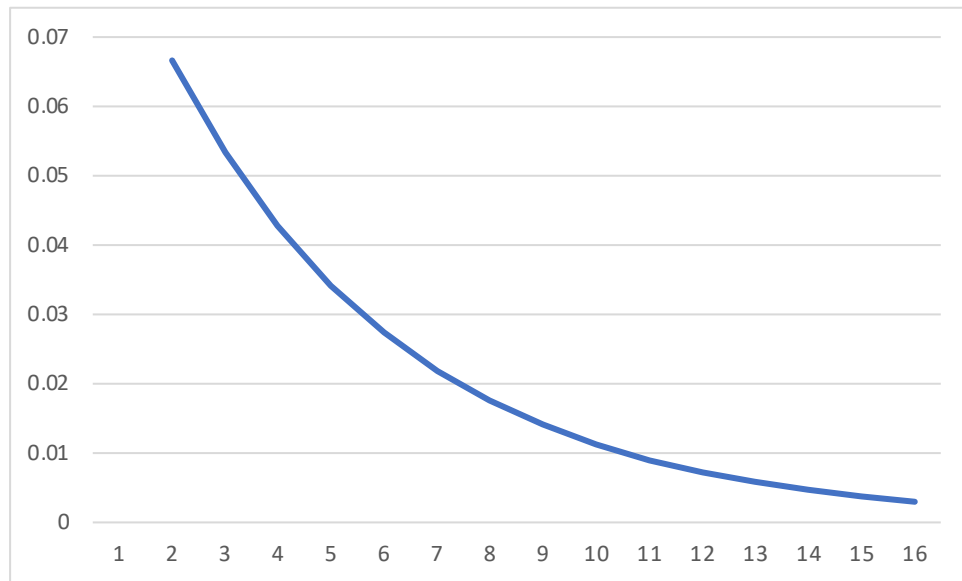


FIGURE 2. CHANGE IN 1-HHI FOR FIRM WITH CONSTANT 20% GROWTH IN NUMBER OF UNITS

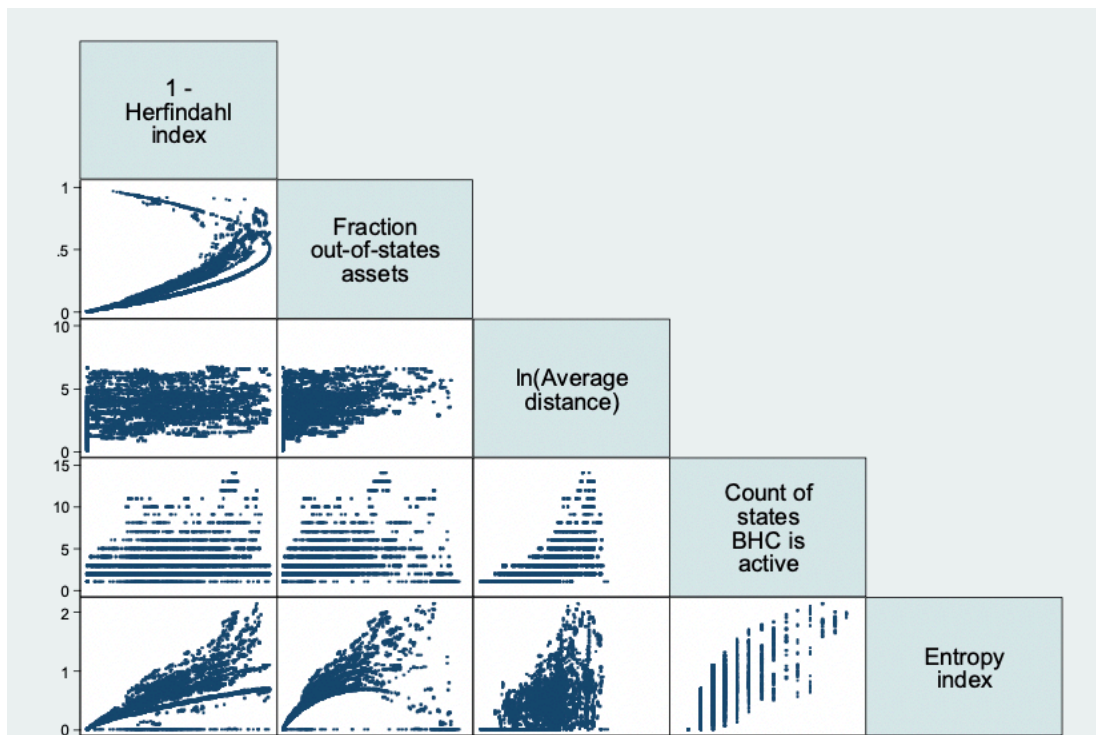


FIGURE 3. CORRELATIONS BETWEEN DIVERSIFICATION MEASURES

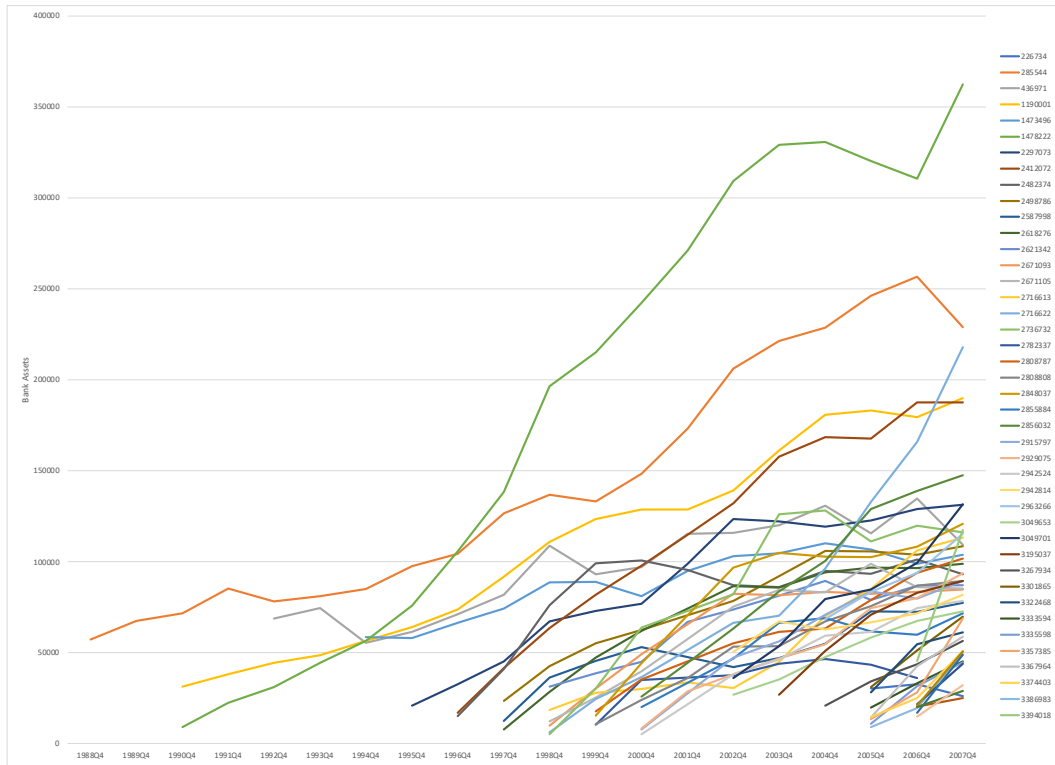


FIGURE 4A. EVOLUTION IN BANK SCALE (TOTAL ASSETS) FOR BHC1247334.

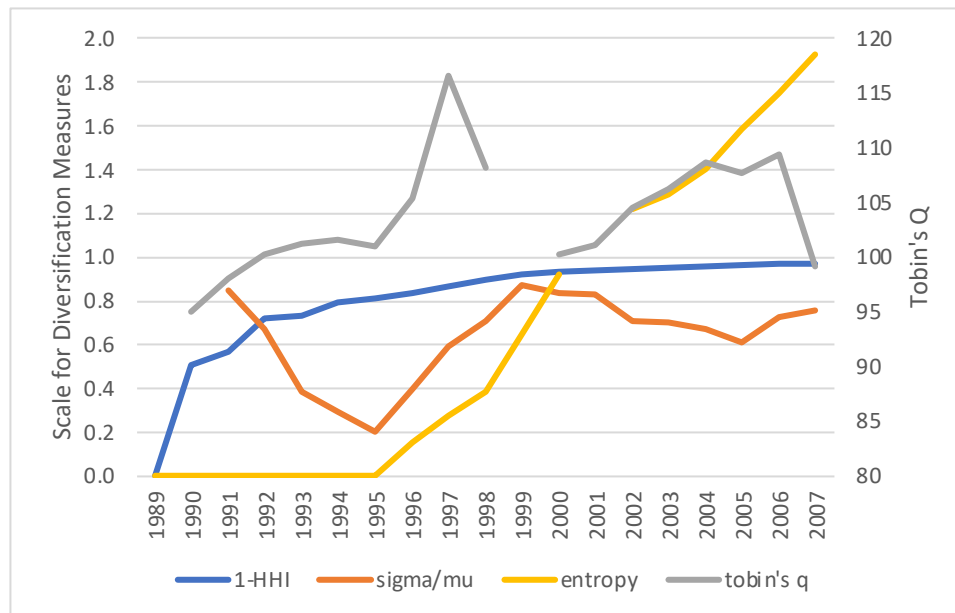


FIGURE 4B. CORRESPONDING EVOLUTION IN MARKET VALUE AND MEASURES OF DIVERSIFICATION

TABLE 1. SUMMARY STATISTICS

Variable	Obs	Mean	StDev	Min	Max
1 - HHI of assets across states	33,211	0.10	0.22	0.00	1.00
Diversification dummy	34,115	0.22	0.41	0.00	1.00
Fraction of assets out-of-state	33,211	0.05	0.13	0.00	0.96
ln(Mean distance between HQ and subsidiaries)	32,414	25.98	71.83	0.06	1040.42
Number of states BHC has subsidiaries	34,132	1.47	1.24	1.00	14.00
Entropy index of assets across states	34,122	0.12	0.28	0.00	2.14
Coefficient of variation (CV) of assets across states	14,351	0.56	0.47	0.00	2.61
Tobin's q	31,847	105.59	5.73	94.70	128.68

TABLE 2. EMPIRICAL ESTIMATION

	Tobin's Q						
	(1)	(2)	(3)	(4)	(5)	(6)	(7)
Diversification dummy	-0.200 (0.334)						
Fraction of assets out-of-state		-0.595 (0.836)					
1 - HHI of assets across states			-0.357 (0.522)				
ln(Mean distance between HQ and subsidiaries)				-0.115 (0.114)			
Number of states BHC has subsidiaries					0.108 (0.089)		
Entropy index of assets across states						0.246 0.414	
Coefficient of variation (CV) of assets across states							-4.67E-07* (2.4E-05)
Observations	31847	33847	31838	31365	31830	31830	6848
R-squared (overall)	0.315	0.316	0.316	0.316	0.321	0.319	0.460
BHC FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Quarter effects	Yes	Yes	Yes	Yes	Yes	Yes	Yes

Robust standard errors (clustered by BHC) in parenthesis

*** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$

TABLE 3. ESTIMATION WITH BALANCED PANEL

	Tobin's Q					
	(1)	(2)	(3)	(4)	(5)	(6)
Fraction of assets out-of-state	2.483*** (1.423)					
1 - HHI of assets across states		2.957*** (1.628)				
ln(Mean distance between HQ and subsidiaries)			-0.003 (0.006)			
Number of states BHC has subsidiaries				0.055 (0.135)		
Entropy index of assets across states					1.216 (0.920)	
Coefficient of variation (CV) of assets across states						-4.66E-07 (4.26E-05)
Observations	1861	1861	1807	1861	1844	1844
R-squared (overall)	0.578	0.577	0.577	0.582	0.585	0.594
BHC FE	Yes	Yes	Yes	Yes	Yes	Yes
Quarter effects	Yes	Yes	Yes	Yes	Yes	Yes

Robust standard errors (clustered by BHC) in parenthesis

*** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$

TABLE 4. DECOMPOSING DIVERSIFICATION INTO COUNT AND SHAPE

	Tobin's Q					
	(1)	(2)	(3)	(4)	(6)	(7)
Number of states BHC has subsidiaries	0.184*	0.205	0.233**	0.186*	0.148	0.125
	0.094	0.110	0.095	0.096	0.133	0.092
Diversification dummy	-0.479					
	0.364					
Fraction of assets out-of-state		-1.546				
		0.999				
1 - HHI of assets across states			-1.020*			
			0.578			
ln(Mean distance between HQ and subsidiaries)				-0.005*		
				(0.003)		
Entropy index of assets across states					-0.232	
					0.666	
Coefficient of variation (CV) of assets across states						-4.8E-05*
						(2.4E-05)
Observations	31846	31838	31838	31364	31829	6848
R-squared (overall)	0.317	0.321	0.320	0.319	0.320	0.464
BHC FE	Yes	Yes	Yes	Yes	Yes	Yes
Quarter effects	Yes	Yes	Yes	Yes	Yes	Yes

Robust standard errors (clustered by BHC) in parenthesis

*** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$

TABLE 5. INSTRUMENTAL VARIABLE REGRESSION

A. STAGE 2

	Tobin's Q					
	(1)	(2)	(3)	(4)	(6)	(7)
Number of states BHC has subsidiaries (IV)	1.330*** (0.521)	2.436** 0.883	3.852*** 1.343	2.466*** 0.742	7.333** 3.489	0.494 0.413
Fraction of assets out-of-state		-11.154*** (4.041)				
1 - HHI of assets across states			-12.059*** (3.811)			
ln(Mean distance between HQ and subsidiaries)				-0.019*** (0.005)		
Entropy index of assets across states					-24.076** (11.277)	
Coefficient of variation (CV) of assets across states						-3.8E-05* (2.4E-05)
Observations	29143	29142	29142	28732	29143	6377
R-squared (overall)	0.306	0.270	0.206	0.264	0.155	0.452
BHC FE	Yes	Yes	Yes	Yes	Yes	Yes
Year effects	Yes	Yes	Yes	Yes	Yes	Yes

Robust standard errors (clustered by BHC) in parenthesis

*** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$

B. STAGE 1

	Number of States with subsidiaries
	(1)
Years since state deregulated	0.139*** (0.032)
(Years since state deregulated)squared	-0.011*** (0.003)
ln(Total operating income)	0.230*** (0.077)
Growth in total operating income	0.249*** (0.077)
Capital asset ratio	0.004 (0.008)
Return on equity	-0.010** (0.005)
Income diversification	-0.992*** (0.252)
Asset Diversification	0.005*** (0.079)
Observations	29818
R-squared (overall)	0.292
BHC FE	Yes
Year effects	No

Robust standard errors (clustered by BHC) in parenthesis

*** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$